ENVS193_Final

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Set Up

Problem 1: Research writing

a) Transparent statistical methods

Part 1: The co-worker likely used a logistic regression because the response variable is the probability of American Avocet microhabitat use (used vs. not used) and the predictor variable is distance to the water edge, which is continuous.

Part 2: The co-worker likely used a chi-square test of independence because both bird species and habitat type are categorical variables, and the goal is to determine whether the distribution of bird species differs among habitat types.

b) Suggestions For Rewriting

Part 1: American Avocets were more likely to use some areas than others depending on their distance from the water's edge, suggesting that proximity to water influences microhabitat selection.

Part 2: The three bird species did not use habitats in the same way. Habitat use differed among species, indicating that each species may prefer different habitat types within the San Francisco Bay.

c) Further Analysis

One additional variable that could influence the probability of American Avocet microhabitat use is water depth (cm), measured at each survey location using a meter stick or depth gauge. Water depth may affect habitat use because American Avocets forage by sweeping their bills through shallow water, so they may prefer areas with depths that maximize feeding efficiency.

Problem 2 Data analysis

a) Clean and Wrangle

```
# Create a new cleaned dataset containing only the variables needed for the nest occurrence a
nests_clean <- nests |>
  # Keep only the response variable, tree height, and ant species columns
  select(case_control, height_cm, ant_species) |>
  # Keep only trees occupied by the three focal ant species
  filter(ant_species %in% c("AB", "BBR", "RRB")) |>
  # Recode and reorder ant species information
  mutate(
    # Convert the ant species codes into a factor and set the desired order
    ant_species = factor(
      ant_species,
      levels = c("BBR", "AB", "RRB"),
      labels = c(
        "C. sjostedti",
        "C. mimosae",
        "C. nigriceps"
      )
    ),
     # Create a new column containing the full scientific name of each ant species
    ant_species_full = case_when(
      ant_species == "C. sjostedti" ~ "Crematogaster sjostedti",
      ant_species == "C. mimosae" ~ "Crematogaster mimosae",
      ant_species == "C. nigriceps" ~ "Crematogaster nigriceps"
    )
  )
# Display the structure of the cleaned dataset to verify variable types and dimensions
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control    : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm       : num [1:270] 392 310 513 154 324 ...
 $ ant_species     : Factor w/ 3 levels "C. sjostedti",..: 3 3 3 3 2 3 1 3 2 1 ...
 $ ant_species_full: chr [1:270] "Crematogaster nigriceps" "Crematogaster nigriceps" "Cremato
```

```
# Display 10 randomly selected rows from the cleaned dataset
nests_clean |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species  ant_species_full
          <dbl>     <dbl> <fct>        <chr>
 1            0       215 C. nigriceps Crematogaster nigriceps
 2            0       163 C. sjostedti Crematogaster sjostedti
 3            1       247 C. nigriceps Crematogaster nigriceps
 4            0       180 C. mimosae   Crematogaster mimosae
 5            0       112 C. sjostedti Crematogaster sjostedti
 6            0       140 C. sjostedti Crematogaster sjostedti
 7            1       234 C. nigriceps Crematogaster nigriceps
 8            0       147 C. nigriceps Crematogaster nigriceps
 9            0       125 C. nigriceps Crematogaster nigriceps
10            0       132 C. nigriceps Crematogaster nigriceps
```

b) Response variable

The response variable is nest occurrence, where a value of 1 indicates that a bird nest was present in a tree and a value of 0 indicates that no bird nest was present. Biologically, this variable represents whether a tree was selected and used as a nesting site by birds.

c) Purpose of Study

Tree height:

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence may increase because taller trees can provide better nesting locations and may be more visible or accessible to birds while also reducing the risk of some predators. This information is discussed in the Methods section, "Nest Site Selection" subsection, where tree characteristics were measured, and is supported by the Results section, where tree height was included as a predictor of nest occurrence.

Acacia ant species:

Acacia ant species is a categorical variable with three levels: C. sjostedti, C. mimosae, and C. nigriceps. The probability of nest occurrence may differ among ant species because birds preferentially nest in trees occupied by more aggressive ant species that defend trees against browsing mammals and potential nest predators. This information is described in the Introduction (final paragraph) and supported in the Discussion section, where the authors explain that birds selected trees inhabited by the more aggressive ant species.

Candidate Logistic Regression Models

Model Name	Response Variable	Predictors
Additive Model	Nest occurrence	Tree height + acacia ant species
Interactive Model	Nest occurrence	Tree height + acacia ant species + tree height x acacia ant spe

d) Table of Models

```r
# Create a table that summarizes the two candidate logistic regression models
model_table <- tibble(
  # Give each model a descriptive name
  `Model Name` = c("Additive Model", "Interactive Model"),
  # Identify the response variable for both models
  `Response Variable` = c(
    "Nest occurrence",
    "Nest occurrence"
  ),
  # Describe the predictors included in each model
  `Predictors` = c(
    "Tree height + acacia ant species",
    "Tree height + acacia ant species + tree height x acacia ant species"
  ),
  # State whether each model includes an interaction term
  `Interaction Term` = c(
    "No",
    "Yes"
  )
)
# Display the model table
model_table |>
  gt() |>
  tab_header(
    title = "Candidate Logistic Regression Models"
  )
```

e) Run the models

```r
# Run the additive logistic regression model
additive_model <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = binomial
)
# Run the interaction logistic regression model
interaction_model <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = binomial
)
```

f) Select the best model

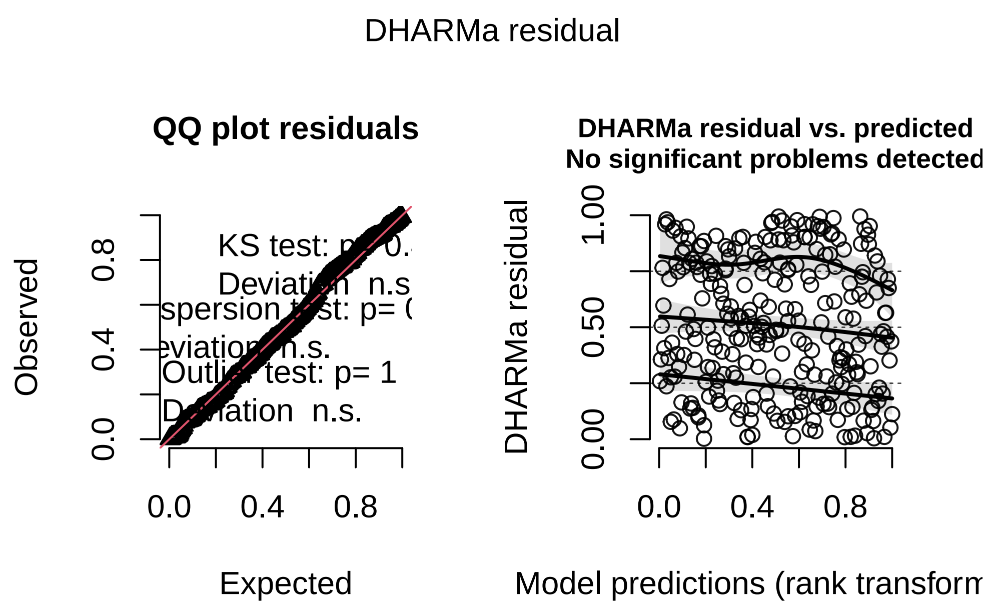
```r
AIC(additive_model, interaction_model)
```

```
                  df      AIC
additive_model     4 258.7430
interaction_model  6 237.8042
```

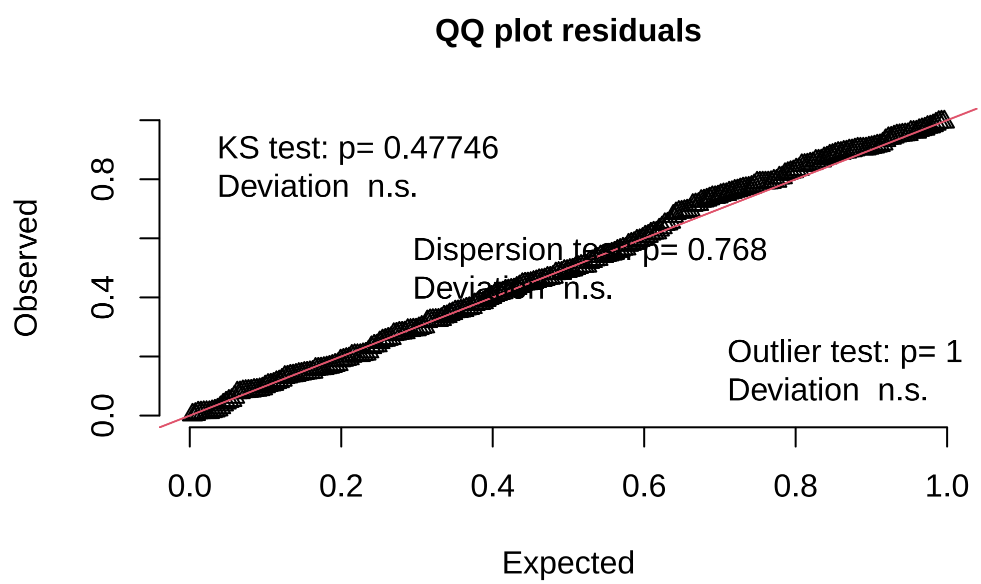
The best model as determined by Akaike's Information Criterion (AIC) was the logistic regression model predicting nest occurrence from tree height, acacia ant species, and the interaction between tree height and acacia ant species. This model had a lower AIC value (237.804) than the model containing only the additive effects of tree height and acacia ant species (258.743), indicating stronger support for the interaction model.

g) Check the diagnostics

```r
# Simulate residuals from the selected model
interaction_residuals <- simulateResiduals(
  fittedModel = interaction_model
)

# Diagnostic plots
plot(interaction_residuals)
```

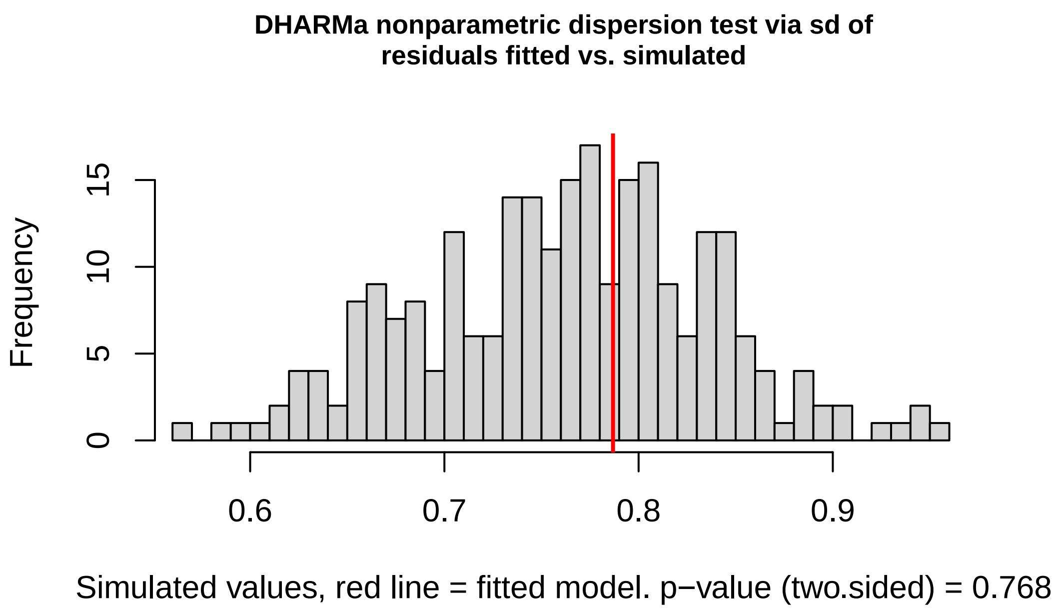



```r
# Test for uniformity
testUniformity(interaction_residuals)
```



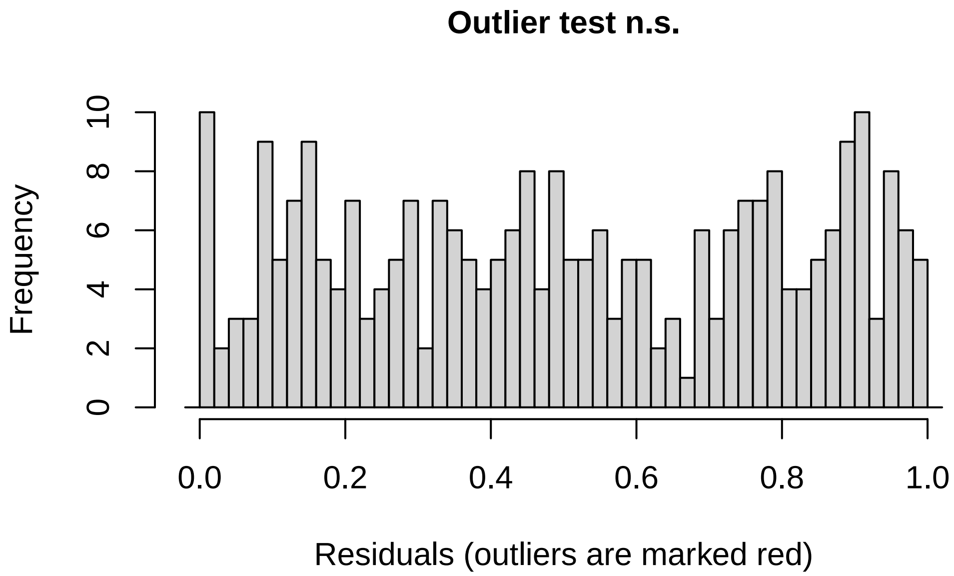
```
	Asymptotic one-sample Kolmogorov-Smirnov test

data:  simulationOutput$scaledResiduals
D = 0.051246, p-value = 0.4775
alternative hypothesis: two-sided
```

```r
# Test for dispersion
testDispersion(interaction_residuals)
```




```
	DHARMa nonparametric dispersion test via sd of residuals fitted vs.
	simulated

data:  simulationOutput
dispersion = 1.0325, p-value = 0.768
alternative hypothesis: two.sided
```

```r
# Test for outliers
testOutliers(interaction_residuals)
```




```
    DHARMa bootstrapped outlier test

data:  interaction_residuals
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
 percent confidence interval:
 0 0
sample estimates:
outlier frequency (expected: 7.40740740740741e-05 )
                                                  0
```

h) Visualize the Model Predictions

Predictions

```
# Create a sequence of tree heights for each ant species
prediction_data <- expand.grid(
  height_cm = seq(
    min(nests_clean$height_cm),
    max(nests_clean$height_cm),
    length.out = 100
```

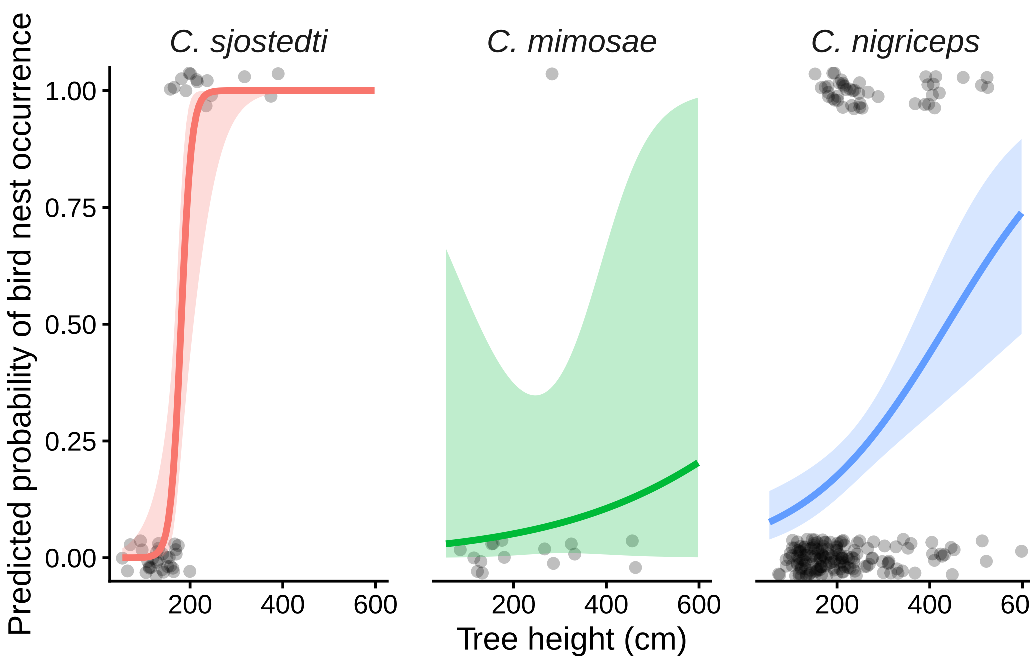
```
  ),
  ant_species = levels(nests_clean$ant_species)
)

# Generate model predictions
predictions <- predict(
  interaction_model,
  newdata = prediction_data,
  type = "link",
  se.fit = TRUE
)

# Calculate predicted probabilities and confidence intervals
prediction_data <- prediction_data |>
  mutate(
    predicted_probability = plogis(predictions$fit),
    lower_ci = plogis(predictions$fit - 1.96 * predictions$se.fit),
    upper_ci = plogis(predictions$fit + 1.96 * predictions$se.fit)
  )
```

Graph

```
# Create a plot showing the observed nest occurrence data and the predicted
# probabilities from the selected interaction model
ggplot() +

# Add the underlying data as jittered points
  geom_jitter(
    data = nests_clean,
    aes(
      x = height_cm,
      y = case_control
    ),
    width = 0,
    height = 0.04,
    alpha = 0.25,
    size = 1.6
  ) +

  # Add shaded 95% confidence intervals around the model predictions
  geom_ribbon(
```

```r
    data = prediction_data,
    aes(
      x = height_cm,
      ymin = lower_ci,
      ymax = upper_ci,
      fill = ant_species
    ),
    alpha = 0.25
  ) +

  # Add the predicted probability lines from the interaction model
  geom_line(
    data = prediction_data,
    aes(
      x = height_cm,
      y = predicted_probability,
      color = ant_species
    ),
    linewidth = 1.2
  ) +

  # Create separate panels for each ant species
  facet_wrap(~ ant_species) +

  # Set y-axis limits and tick marks to represent probabilities
  scale_y_continuous(
  breaks = seq(0, 1, 0.25)
) +
coord_cartesian(
  ylim = c(0, 1)
) +

  # Add informative axis labels
  labs(
    x = "Tree height (cm)",
    y = "Predicted probability of bird nest occurrence"
  ) +

  # Use a clean, non-default theme
  theme_classic() +

  # Remove the legend and improve readability
```

```r
  theme(
    legend.position = "none",
    strip.background = element_blank(),
    strip.text = element_text(
      size = 12,
      face = "italic"
    ),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10),
    panel.spacing = unit(1.2, "lines")
  )
```



i) Write a Caption

Figure 1. Predicted probability of bird nest occurrence by tree height and acacia ant species. Points represent individual trees, where 0 indicates no bird nest was present and 1 indicates a bird nest was present. Colored lines show the predicted probability of nest occurrence from the selected logistic regression model, and shaded ribbons represent 95% confidence intervals around the predictions. Separate panels display predictions for trees occupied by Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. Data source: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for:

Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j) Calculate Model Predictions

```r
# Create a new data frame containing a tree height of 600 cm for each of the
# three acacia ant species
prediction_600 <- tibble(
  height_cm = 600,
  ant_species = factor(
    c(
      "C. sjostedti",
      "C. mimosae",
      "C. nigriceps"
    ),
    levels = levels(nests_clean$ant_species)
  )
)

# Calculate predicted probabilities using the selected interaction model
prediction_600 <- prediction_600 |>
  mutate(
    predicted_probability = predict(
      interaction_model,
      newdata = prediction_600,
      type = "response"
    )
  )

# Display the predicted probabilities
prediction_600
```

```
# A tibble: 3 x 3
  height_cm ant_species  predicted_probability
      <dbl> <fct>                        <dbl>
1       600 C. sjostedti                 1
2       600 C. mimosae                   0.205
3       600 C. nigriceps                 0.741
```

k) interpret results

The probability of bird nest occurrence differed among acacia ant species and changed with tree height (Figure 1). At the tallest observed tree height of 600 cm, the predicted probability of nest occurrence was 1.00 for trees occupied by C. sjostedti, 0.741 for trees occupied by C. nigriceps, and 0.205 for trees occupied by C. mimosae. Figure 1 shows that the relationship between tree height and nest occurrence depended on ant species, which is consistent with the interaction model being selected by AIC. Birds may preferentially nest in trees occupied by certain acacia ant species because aggressive ant defenders can help protect nests from predators and herbivores, making those trees safer nesting locations.

Problem 3

a) Comparing Visualizations

- My exploratory visualizations represented the data quantitatively using axes, points, and summary statistics to show relationships between variables such as workout type and gym time. In contrast, my affective visualization represented the data metaphorically as a mountain landscape, where peaks and valleys reflected changes in gym attendance and workout duration over time. The affective visualization focused more on communicating the experience of the data than on precise numerical values.
- All of the visualizations were based on the same gym time dataset and communicated patterns in how often and how long I worked out. Each visualization highlighted variation in gym attendance across days and emphasized periods of higher and lower activity. Although the visual styles differed, they all represented the same underlying trends in my workout behavior.
- The exploratory visualizations showed clear differences in gym time among workout types and revealed day-to-day variation in workout duration. The affective visualization showed similar patterns through changes in mountain height, with taller peaks corresponding to longer workouts and valleys representing rest days. The overall patterns were similar because they were based on the same data, but the affective visualization emphasized the emotional experience of consistency and effort rather than precise statistical comparisons.
- During workshop, I received feedback to make the connection between the mountain landscape and the gym data more obvious and to simplify some of the visual elements. I implemented these suggestions by making the peaks correspond more directly to workout duration and reducing unnecessary background details. These changes helped the visualization communicate the data more clearly while still maintaining its artistic and affective style

b) Designing an analysis

- To answer my research question about how workout type influences the amount of time I spend at the gym, I would use a one-way ANOVA. This analysis is appropriate because my response variable, gym time (minutes), is continuous, while my predictor variable, workout type, is categorical with three groups (Push, Pull, and Legs). A one-way ANOVA allows me to test whether the mean gym time differs among the three workout types and determine whether certain workouts tend to require more time than others.

c) Check Assumptions and Run Test

```r
# Remove days without a workout because N/A is not a workout type
gym_clean <- gym_data |>
  filter(`Workout Type` %in% c("Push", "Pull", "Leg"))

# Run a one-way ANOVA testing whether mean gym time differs
# among Push, Pull, and Leg workouts
gym_anova <- aov(
  `Gym Time (min)` ~ `Workout Type`,
  data = gym_clean
)

# Display ANOVA results
summary(gym_anova)
```

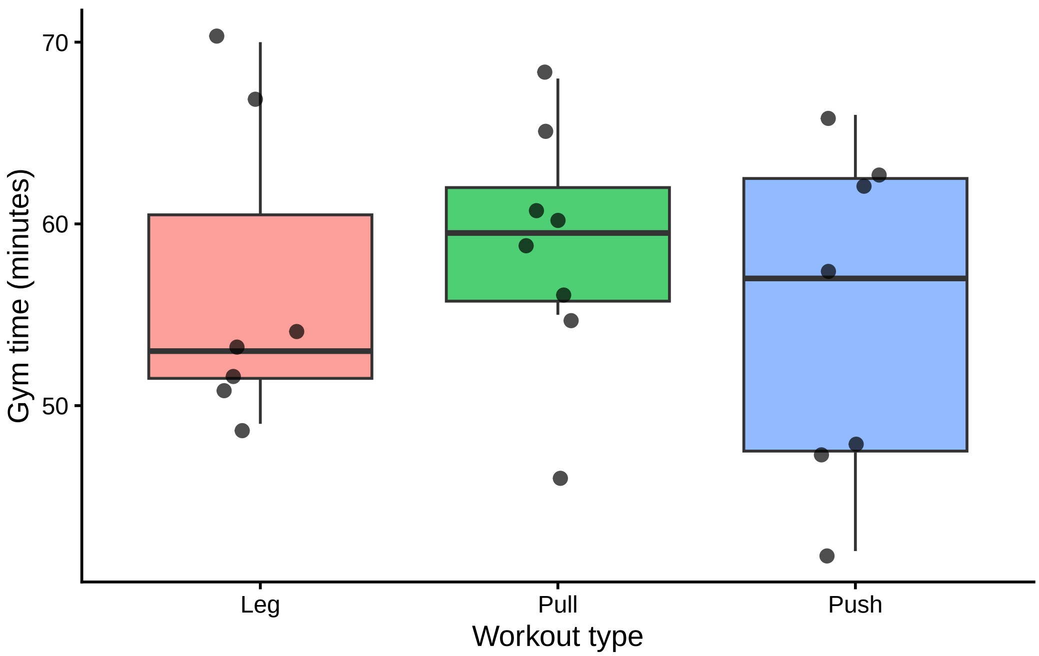
```
               Df Sum Sq Mean Sq F value Pr(>F)
`Workout Type`  2   53.4   26.69   0.405  0.673
Residuals      19 1253.2   65.96
```

```r
# Calculate eta-squared effect size
anova_table <- summary(gym_anova)[[1]]

eta_sq <- anova_table$`Sum Sq`[1] /
  sum(anova_table$`Sum Sq`)

eta_sq
```

```
[1] 0.04085183
```

d) Visualization

```
ggplot(
  gym_clean,
  aes(
    x = `Workout Type`,
    y = `Gym Time (min)`,
    fill = `Workout Type`
  )
) +
  geom_boxplot(
    alpha = 0.7,
    outlier.shape = NA
  ) +
  geom_jitter(
    width = 0.15,
    alpha = 0.7,
    size = 2
  ) +
  labs(
    x = "Workout type",
    y = "Gym time (minutes)"
  ) +
  theme_classic() +
  theme(
    legend.position = "none"
  )
```



e) Figure caption

- Figure 1. Gym time by workout type. Points represent individual gym sessions and show the amount of time spent at the gym for Push, Pull, and Leg workouts. Black points represent the mean gym time for each workout type. Colors distinguish workout types, and the plot displays the underlying data used in the one-way ANOVA.

f) Write About Results

- Workout type did not have a significant effect on gym time (one-way ANOVA: $F(2, 19) = 0.405$, $p = 0.673$). The effect size was small ($^2 = 0.041$), indicating that workout type explained only about 4.1% of the variation in gym time. Mean gym time was similar among Push, Pull, and Leg workouts, suggesting that the type of workout performed did not strongly influence how long I spent at the gym. Other factors, such as available free time, schedule constraints, or gym crowdedness, may have had a greater influence on workout duration.